An Analysis of Stacking ensemble

Stacking ensemble

Stacking ensemble -- Part 1

where y {\displaystyle y} is the predicted class, C {\displaystyle C} is the set of all possible classes, H {\displaystyle H} is the hypothesis space, P {\displaystyle P} refers to a probability, and T {\displaystyle T} is the training data. As an ensemble, the Bayes optimal classifier represents a hypothesis that is not necessarily in H {\displaystyle H} . The hypothesis represented by the Bayes optimal classifier, however, is the optimal hypothesis in ensemble space (the space of all possible ensembles consisting only of hypotheses in H {\displaystyle H}). This formula can be restated using Bayes' theorem, which says that the posterior is proportional to the likelihood times the prior:

Stacking ensemble -- Part 2

Bucket of models A "bucket of models" is an ensemble technique in which a model selection algorithm is used to choose the best model for each problem. When tested with only one problem, a bucket of models can produce no better results than the best model in the set, but when evaluated across many problems, it will typically produce much better results, on average, than any model in the set. The most common approach used for model-selection is cross-validation selection (sometimes called a "bake-off contest"). It is described with the following pseudo-code:

Stacking ensemble -- Part 3

The geometric framework Ensemble learning, including both regression and classification tasks, can be explained using a geometric framework. Within this framework, the output of each individual classifier or regressor for the entire dataset can be viewed as a point in a multi-dimensional space. Additionally, the target result is also represented as a point in this space, referred to as the "ideal point." The Euclidean distance is used as the metric to measure both the performance of a single classifier or regressor (the distance between its point and the ideal point) and the dissimilarity between two classifiers or regressors (the distance between their respective points). This perspective transforms ensemble learning into a deterministic problem. For example, within this geometric framework, it can be proved that the averaging of the outputs (scores) of all base classifiers or regressors can lead to equal or better results than the average of all the individual models. It can also be proved that if the optimal weighting scheme is used, then a weighted averaging approach can outperform any of the individual classifiers or regressors that make up the ensemble or as good as the best performer at least.

Stacking ensemble -- Part 4

y = a r g m a x c j ∈ C ∑ h i ∈ H P (c j | h i) P (h i | T) {\displaystyle y={\underset {c_{j}\in C}{\mathrm {argmax} }}\sum _{h_{i}\in H}{P(c_{j}|h_{i})P(h_{i}|T)}}

Stacking ensemble -- Part 5

Ensemble learning applications In recent years, due to growing computational power, which allows for training in large ensemble learning in a reasonable time frame, the number of ensemble learning applications has grown increasingly. Some of the applications of ensemble classifiers include:

Stacking ensemble -- Part 6

Financial decision-making Ensemble learning methods have been widely adopted in finance for tasks such as credit scoring, bankruptcy prediction, and risk management. By combining multiple base models, ensembles can exploit nonlinear relationships, handle high-dimensional and noisy data, and often deliver more stable out-of-sample performance than single models or traditional statistical baselines such as logistic regression and linear factor models. This approach aligns with the broader trend in financial machine learning described by Marcos López de Prado, who argues that combining diverse learners can improve robustness, mitigate overfitting, and extract persistent patterns from noisy financial time series. In retail and corporate credit scoring, ensemble classifiers such as random forests, gradient boosting machines and stacked models are frequently used to assess default risk. A recent systematic literature review of machine-learning credit-scoring studies published between 2018 and 2024 reports that tree-based ensembles and boosting methods are among the most commonly applied techniques and typically achieve higher predictive accuracy than traditional scorecards or single classifiers. Ensemble methods have also been applied to corporate failure and bankruptcy prediction. Studies comparing different ensemble constructions (such as bagging, boosting and heterogeneous classifier pools) report that well-tuned ensembles tend to achieve higher classification accuracy and more robust performance across industries than individual models. The accuracy of prediction of business failure is a very crucial issue in financial decision-making. Therefore, different ensemble classifiers are proposed to predict financial crises and financial distress. Also, in the trade-based manipulation problem, where traders attempt to manipulate stock prices by buying and selling activities, ensemble classifiers are required to analyze the changes in the stock market data and detect suspicious symptom of stock price manipulation. Implementation of ensemble learning not only with machine learning models but also with mathematical frameworks such as hidden Markov models for applications in trading, regime detection, and asset pricing remains a rapidly developing area of research.

Stacking ensemble -- Part 7

Medicine Ensemble classifiers have been successfully applied in neuroscience, proteomics and medical diagnosis like in neuro-cognitive disorder (i.e. Alzheimer or myotonic dystrophy) detection based on MRI datasets, cervical cytology classification. Besides, ensembles have been successfully applied in medical segmentation tasks, for example brain tumor and hyperintensities segmentation.

Stacking ensemble -- Part 8

For each model m in the bucket: Do c times: (where 'c' is some constant) Randomly divide the training dataset into two sets: A and B Train m with A Test m with B Select the model that obtains the highest average score

Stacking ensemble -- Part 9

e k = H (p , q k) − λ K ∑ j ≠ k H (q j , q k) {\displaystyle e^{k}=H(p,q^{k})-{\frac {\lambda }{K}}\sum _{j\neq k}H(q^{j},q^{k})}

Stacking ensemble -- Part 10

Amended Cross-Entropy Cost: An Approach for Encouraging Diversity in Classification Ensemble The most common approach for training classifier is using Cross-entropy cost function. However, one would like to train an ensemble of models that have diversity so when we combine them it would provide best results. Assuming we use a simple ensemble of averaging K {\displaystyle K} classifiers. Then the Amended Cross-Entropy Cost is

Stacking ensemble -- Part 11

where e k {\displaystyle e^{k}} is the cost function of the k t h {\displaystyle k^{th}} classifier, q k {\displaystyle q^{k}} is the probability of the k t h {\displaystyle k^{th}} classifier, p {\displaystyle p} is the true probability that we need to estimate and λ {\displaystyle \lambda } is a parameter between 0 and 1 that define the diversity that we would like to establish. When λ = 0 {\displaystyle \lambda =0} we want each classifier to do its best regardless of the ensemble and when λ = 1 {\displaystyle \lambda =1} we would like the classifier to be as diverse as possible.